

Unit III 3 mark question and answer

1. What is primary data?

Primary Data is the data that is collected first-hand by the researcher directly from original sources for a specific research purpose. It is fresh, original, and has not been previously published or used.

Characteristics of Primary Data:

- Collected directly from respondents or real-world observations.
- Specific to the problem under study.
- More reliable but time-consuming and costly to gather.

Examples:

- Surveys and questionnaires
- Interviews
- Experiments
- Direct observations
- Focus group discussion

2. Differentiate between primary and secondary data

Basis	Primary Data	Secondary Data
Definition	First-hand data collected directly by the researcher for a specific purpose.	Data already collected and recorded by others for different purposes, but used by the researcher.
Source	Surveys, interviews, experiments, observations, questionnaires, focus groups	Books, journals, newspapers, government reports, company records, websites, databases.
Nature	Original, specific, and up-to-date	Already existing, may be outdated or not specific
Cost	Expensive to collect (time, money, effort)	Relatively cheaper since data already exists
Reliability	Generally more reliable since collected directly	Reliability depends on the original source and method of collection.
Time consuming	Time Consuming	Less time-consuming
Purpose	Collected for the specific research problem at hand.	Collected for purposes other than the current research.

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3. Define the term “sampling.”

Sampling is the process of selecting a small group of individuals, items, or observations (called a sample) from a larger population in such a way that the selected group represents the characteristics of the whole population.

4. What is a research instrument?

A research instrument is a tool or device used by researchers to collect, measure, and analyze data related to a research problem.

Examples

- Questionnaires
- Interviews (structured/unstructured)
- Observation checklists
- Surveys
- Tests or scales

5. Explain the significance of sampling in research

1. Cost-effectiveness

Studying the entire population (census) is expensive. Sampling reduces costs by focusing only on a smaller group.

2. Saves Time and Effort

Collecting data from a sample is quicker and more practical than covering the whole population.

3. Feasibility

In many cases, studying the whole population is not possible.

4. Better data quality

Studying the entire population (census) is expensive. Sampling reduces costs by focusing only on a smaller group.

5. Basis for Generalization

Proper sampling techniques ensure that results from the sample can be generalized to represent the entire population.

6.Focus on specific Groups

Sampling allows researchers to study particular segments of the population in depth.

6.What is stratified sampling?

Stratified Sampling is a probability sampling technique in which the population is divided into different subgroups (strata) based on specific characteristics such as age, gender, income, education, etc., and then samples are drawn from each subgroup.

Types

- Proportionate stratified sampling
- Disproportionate stratified sampling

7.Describe the meaning of non-probability sampling

Non-Probability Sampling is a sampling technique in which not all members of the population have a chance of being selected. The selection of samples is based on the judgment, convenience, or purpose of the researcher, rather than random selection.

Key Points:

- Does not rely on chance or randomness
- Easier, quicker, and less costly compared to probability sampling.
- Results cannot always be generalized to the whole population.

8.What is meant by validity in research instruments?

Validity in research instruments refers to the degree to which an instrument accurately measures what it is intended to measure. It shows whether the data collected truly represents the concept under study.

Key Points:

- A valid instrument ensures accuracy and relevance of results.
- Without validity, even reliable instruments may give misleading findings.
- It checks whether the instrument is measuring the right thing, not just giving consistent results

9.What is reliability, and why is it important

Reliability in research refers to the consistency or stability of a research instrument in measuring something. An instrument is said to be reliable if it gives the same results when repeated under similar conditions.

Importance of Reliability:

1. Consistency of Results

Ensures that findings are dependable and not due to random errors.

2. Trustworthiness

Increases the credibility of research outcomes.

3. Reproducibility

Other researchers can repeat the study and obtain similar results.

4. Foundation for Validity

A tool must be reliable before it can be valid.

10. List different data collection methods

1. Observation Method

Collecting data by directly watching people, events, or behaviors.

2. Interview Method

Collecting information through face-to-face, telephone, or online interviews (structured or unstructured).

3. Questionnaires/Surveys

Distributing written or online sets of questions to respondents.

4. Experiments

Collecting data by conducting controlled experiments to observe cause-and-effect relationships.

5. Focus Groups

Gathering data through group discussions on a specific topic.

6. Case Studies

In-depth investigation of a single person, group, or situation.

7. Secondary Data Method

Using already available information from books, reports, articles, databases, or government records.

8. Online/ Digital Data Collection

Using online tools, social media, or digital platforms to gather data.

11. What are the characteristics of a good sample design?

Characteristics of a Good Sample Design

1. Representativeness

The sample should closely resemble the characteristics of the entire population.

2. Accuracy

It should minimize sampling errors and provide reliable and valid results.

3. Adequate Size

The sample must be large enough to give dependable results but not unnecessarily large.

4. Practicality

The design should be simple, economical, and feasible to implement.

5. Flexibility

It should allow adjustments if conditions change during research.

6. Efficiency

The design should give maximum information with minimum effort, time, and cost.

7. Objectivity

The selection of the sample should be free from bias and based on scientific methods.

8. Reliability

The results obtained from the sample should be consistent when the study is repeated.

12. Define systematic sampling with an example

Systematic Sampling is a probability sampling technique in which the first unit is selected at random, and then every kth unit from the population is chosen for the sample.

Here, k is called the sampling interval and is calculated as:

$$k = \frac{\text{Population Size (N)}}{\text{Sample Size (n)}}$$

Example:

Suppose a researcher wants to select a sample of 100 students from a population of 1,000 students.

Sampling interval.

If the first student is chosen randomly as the 5th student, then the sample will include the 5th, 15th, 25th, 35th ... and so on until 100 students are selected.

13. What is a Likert scale?

Likert Scale is a psychometric scale commonly used in research questionnaires to measure attitudes, opinions, or perceptions of respondents. It presents a statement and asks respondents to indicate their level of agreement or disagreement on a graded scale.

Key Points:

- Developed by Rensis Likert in 1932.
- Usually ranges from 5-point to 7-point scales (e.g., Strongly Agree → Strongly Disagree).
- Provides quantitative data for qualitative concepts like satisfaction, attitude, or opinion.
- Easy to construct and widely used in social sciences, marketing, and psychology.

Example (5-point Likert scale):

- Strongly Agree
- Agree

- Neutral
- Disagree
- Strongly Disagree

14. Describe the difference between random sampling and convenience sampling

Basis	Random Sampling	Convenience Sampling
Definition	A probability sampling method where every member of the population has an equal chance of being selected	A non-probability sampling method where samples are selected based on ease of access and availability.
Selection Method	Uses random techniques (lottery, random numbers). Based on researcher's convenience	Bias Minimizes bias since selection is random. High risk of bias as selection is not random.
Representativeness	More representative of the whole population	Less representative, may not reflect the population accurately.
Accuracy & Reliability	High accuracy and reliability.	Lower accuracy, results may not be reliable.

15. What is the purpose of a pilot study?

Purpose of a Pilot Study

A pilot study is a small-scale preliminary study conducted before the main research. Its purpose is to test the feasibility, time, cost, methods, and instruments to be used in the actual research.

Key Purposes:

- Feasibility Check
- Testing Research Instruments

- Identify Problems
- Refining Methods
- Time and Cost Estimation
- Training for Researchers
- Enhance Reliability and Validity

16. Define longitudinal research.

Longitudinal Research is a type of research design in which data is collected from the same subjects repeatedly over a long period of time to study changes, trends, or developments.

Key Points:

- Focuses on studying change over time.
- Involves repeated observations, surveys, or measurements.
- Useful in fields like psychology, education, medicine, and social sciences.
- Helps in identifying cause-and-effect relationships and long-term patterns.

17. What is a research population?

Research Population refers to the entire group of individuals, items, or elements that share common characteristics and are of interest to the researcher. It is the whole set from which a sample may be drawn for a study.

Key Points:

- It includes all possible subjects relevant to the research problem.
- Can be finite (e.g., students in a college) or infinite (e.g., all stars in the sky).
- The population is the target group, while **a sample is a smaller group selected from it.**

18. How is a research sample selected?

A research sample is selected through a systematic process

1. Define the Population

Clearly identify the group from which the sample will be drawn.

2. Decide the Sample Size

Determine how many individuals or elements should be included.

3. Choose a Sampling Method

Select an appropriate technique based on the research type:

- Probability Sampling
- Non-Probability Sampling

4. Select the Sample

Apply the chosen method to pick the required number of participants.

5. Check for Representativeness

Ensure the sample reflects the population's key characteristics .

19.Explain the purpose of random sampling.

Random sampling is a probability sampling method where every member of the population has an equal chance of being selected. Its main purpose is to make research findings more accurate and unbiased.

Key Purposes:

1. Eliminate Bias

Ensures that the researcher's personal choice does not influence sample selection.

2. Representativeness

Provides a fair representation of the whole population.

3. Accuracy of Results

Increases the reliability and validity of conclusions.

4. Generalization

Makes it possible to generalize the findings from the sample to the entire population.

5. Simplicity

The method is straightforward and easy to apply using random numbers or software.

6. Scientific Rigor

Adds credibility and objectivity to the research process.
20. Why is a control group significant in experimental research? A control group is the group in an experiment that does not receive the treatment or intervention being tested. It is kept under normal conditions and is compared with the experimental group. <u>Importance:</u> 1. Basis for Comparison Provides a standard to compare the effects of the treatment. 2. Eliminates Alternative Explanations Helps ensure that changes in the experimental group are due to the treatment, not other factors. 3. Increases Validity Improves the internal validity of the research by controlling extraneous variables.

Unit IV 3 marks question and answer

1. What is data analysis?

Data Analysis is the process of systematically examining, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making.

It involves identifying patterns, relationships, and trends within data so that organizations or researchers can make informed choices.

Key steps in data analysis include:

- Data Collection
- Data Cleaning
- Data Processing & Transformation –
- Exploratory Analysis
- Statistical/Computational Analysis
- Interpretation & Visualization

2. What are the approaches, types of data and measures of distance used in cluster analysis?

1.Approaches to Cluster Analysis

Partitioning Methods:

Divides data into k groups (e.g., k-means, k-medoids).

Hierarchical Methods:

Creates a tree-like structure of clusters (e.g., agglomerative, divisive).

Density-Based Methods:

Groups data points based on dense regions in the data space (e.g., DBSCAN).

Grid-Based Methods:

Divides the data space into a finite number of cells and clusters them (e.g., STING).

Model-Based Methods:

Assumes a model for each cluster and fits the data accordingly (e.g., Gaussian Mixture Models).

2. Types of Data in Cluster Analysis

- Interval-Scaled Data
- Binary Data
- Nominal Data
- Ordinal Data
- Ratio-Scaled Data
- Mixed-Type Data

3. Measures of Distance (Similarity/Dissimilarity)

- Euclidean Distance
- Manhattan Distance (City Block / L1 Norm)
- Minkowski Distance
- Chebyshev Distance
- Cosine Similarity
- Jaccard Coefficient
- Mahalanobis Distance

3. What are parametric tests?

Parametric Tests are statistical tests that make specific assumptions about the population from which the samples are drawn.

These assumptions usually include that the data:

1. Follows a normal distribution (or approximately normal).
2. Is measured on an interval or ratio scale.
3. Has homogeneity of variance (equal variances across groups).

Examples of Parametric Tests:

- t-test (one-sample, independent, paired)
- Analysis of Variance (ANOVA)
- Z-test
- Pearson's correlation coefficient

4. Define the term "correlation."

- Correlation is a statistical measure that describes the degree and direction of the relationship between two variables.
- If one variable changes and the other tends to change in a predictable way, they are said to be correlated.
- Correlation can be positive (both increase together), negative (one increases while the other decreases), or zero (no relationship).

It is commonly measured using the correlation coefficient which ranges from -1 to $+1$

5. What is regression analysis?

Regression Analysis is a statistical method used to study the relationship between a dependent variable (outcome) and one or more independent variables (predictors).

It helps to estimate how the dependent variable changes when the independent variable(s) change.

It is mainly used for prediction, forecasting, and identifying cause-effect relationships.

Types of Regression Analysis:

- Simple Linear Regression
- Multiple Regression
- Non-linear Regression

6. Describe the purpose of editing data in research

Editing of data in research is the process of reviewing and checking the collected data to identify and correct errors, omissions, inconsistencies, and inaccuracies before analysis.

Purpose of Editing Data:

1. Ensure Accuracy

To detect and correct mistakes made during data collection.

2. Maintain Consistency

To standardize responses and avoid contradictions.

3. Improve Completeness

To check for missing data and fill gaps if possible.

4. Enhance Reliability

To make the data dependable for further analysis.

5. Facilitate Coding & Analysis

To organize data in a clear, error-free form for statistical processing.

7. Why is coding data important?

Coding of data is the process of assigning symbols, numbers, or categories to responses or pieces of information so that they can be systematically organized and analyzed.

Importance of Coding Data:

1. Simplifies Data

Converts raw data into a manageable form.

2.Enables Statistical Analysis

Makes it possible to apply mathematical and statistical techniques.

3.Improves Accuracy

Reduces errors in handling large volumes of data.

4.Facilitates Comparison

Allows easy comparison between different groups or variables.

5.Saves Time

Speeds up data entry, processing, and interpretation.

6.Ensures Consistency

Provides uniformity in categorizing responses.

8.Define measures of central tendency

Measures of Central Tendency are statistical values that describe the center or average of a data set. They indicate a typical or representative value around which the data tends to cluster.

Main Measures of Central Tendency:

Mean

The arithmetic average of all values.

Median

The middle value when the data is arranged in order.

Mode

The value that occurs most frequently.

9.How does discriminant analysis work and when to use it?

How it Works:

Data Input

Collect data with known group memberships (e.g., customers who bought vs. did not buy a product).

Function Creation

The method develops a discriminant function using predictor variables.

Classification

New cases are placed into the group whose mean is closest, based on the discriminant score.

Validation

Accuracy of classification is tested.

When to Use Discriminant Analysis:

- When the dependent variable is categorical (e.g., yes/no, type of customer).
- When the independent variables are interval or ratio-scaled (e.g., income, age, test scores).
- For classification and prediction (e.g., credit risk assessment, medical diagnosis, customer segmentation).

10.Explain what is meant by a Chi-square test.

Chi-square test is a non-parametric statistical test used to determine whether there is a significant association or difference between observed frequencies (actual data) and expected frequencies (theoretical data).

Key Points:

- It compares what we actually observe in the data with what we would expect if there were no relationship or difference.
- It is mainly used for categorical data (nominal or ordinal).
- The test statistic follows a Chi-square (χ^2) distribution.

Types of Chi-square Tests:

- Chi-square Test of Independence
- Chi-square Goodness-of-Fit Test

11.What is ANOVA?

ANOVA (Analysis of Variance) is a statistical technique used to test whether there are significant differences between the means of three or more groups.

Key Points:

- It compares the variation between group means with the variation within groups.
- If the between-group variation is much larger than the within-group variation, it indicates that at least one group mean is significantly different.
- It assumes data is normally distributed and variances are equal across groups.

Types of ANOVA:

1. One-way ANOVA
2. Two-way ANOVA
3. MANOVA (Multivariate ANOVA)

12. Define skewness. Compare zero skewness, Right skewness and Left skewness.

Skewness is a statistical measure that describes the asymmetry of the distribution of data around its mean.

A distribution may be symmetric (zero skewness) or asymmetric (positive or negative skewness).

Types of Skewness:

Zero Skewness (Symmetrical Distribution):

The left and right sides of the distribution are mirror images.

Mean = Median = Mode.

Right Skewness (Positively Skewed):

The tail of the distribution extends more to the right.

Mean > Median > Mode.

Left Skewness (Negatively Skewed):

The tail of the distribution extends more to the left.

Mean < Median < Mode.

13.What is factor analysis used for?

Factor Analysis is a statistical technique used to identify the underlying factors (latent variables) that explain the patterns of correlations among a large set of observed variables.

Uses of Factor Analysis:

Data Reduction

Reduces a large number of variables into fewer meaningful factors.

Identify SStructure

Helps to find hidden relationships between variables.

Simplify Complex Data

Makes data easier to interpret by grouping related variables.

Remove Multicollinearity

Helps in regression analysis by reducing highly correlated variables.

14.Define multivariate analysis

Multivariate Analysis is a set of statistical techniques used to analyze data that involves two or more variables at the same time.

It helps in understanding the relationships, patterns, and effects among multiple dependent and independent variables simultaneously.

Key Points:

- Deals with complex datasets involving several variables.
- Used for prediction, classification, data reduction, and hypothesis testing.

15.Why is statistical software important in data analysis?

Statistical software is important in data analysis because it provides tools to efficiently handle, process, and analyze large amounts of data with accuracy and speed.

Importance of Statistical Software in Data Analysis:

Accuracy

Minimizes human errors in complex calculations.

Efficiency

Saves time by automating data processing and statistical tests.

Handles Large Data

Can process and analyze massive datasets that manual methods cannot.

Advanced Techniques

Provides access to sophisticated statistical models and methods (e.g., regression, factor analysis, ANOVA).

Visualization

Generates graphs, charts, and plots for better understanding of results.

Reproducibility

Ensures results can be replicated easily.

User-Friendly

Makes analysis easier for researchers without deep mathematical background.

16. Define a research question.

Research Question is a clear, focused, and specific question that a research study seeks to answer. It defines the problem to be investigated and guides the direction of the research process.

Key Points:

- It identifies what the researcher wants to know.
- It should be clear, concise, and researchable.
- It helps in choosing the research design, data collection, and analysis methods.

17. What is statistical data analysis?

Statistical Data Analysis is the process of applying statistical methods and techniques to collect, organize, summarize, interpret, and draw conclusions from data.

Key Points:

- It helps to identify patterns, relationships, and trends in data.
- Uses descriptive statistics (mean, median, mode, standard deviation) to summarize data.
- Uses inferential statistics (hypothesis testing, regression, ANOVA, chi-square) to make predictions or generalizations about a population from a sample.
- Supports decision-making, forecasting, and problem-solving.

18.Explain data processing and analysis methods.

Data processing is the process of converting raw data into a usable format, while data analysis involves examining that processed data to extract meaningful insights. Both are crucial steps in research and decision-making.

Data Processing Methods:

Editing

Checking data for errors, omissions, and inconsistencies.

Coding

Assigning symbols or numbers to responses for easier handling.

Classification

Organizing data into categories or groups based on characteristics.

Tabulation

Arranging data in rows and columns to summarize information clearly.

Data Analysis Methods:

Descriptive Analysis

- Summarizes and describes data using:
- Measures of central tendency (mean, median, mode)
- Measures of dispersion (range, variance, standard deviation)
- Frequency distributions, charts, and graphs

Inferential Analysis

Makes predictions or generalizations about a population based on sample data using:

Visualization

Uses graphs, charts, and plots to present data insights clearly.

19. List different statistical tests for hypothesis testing.

1. Parametric Tests

- It assumes normal distribution and interval/ratio data.
- t-test – Compares means of two groups
- One-sample t-test
- Independent (two-sample) t-test
- Paired t-test

2. Non-Parametric Tests

- It is used when data doesn't meet parametric assumptions or is ordinal/nominal.
- Chi-square test – Tests association between categorical variables
- Test of independence
- Goodness-of-fit test

3. Multivariate Tests

- It is for multiple variables.
- Discriminant Analysis – Classifies cases into groups
- Factor Analysis – Reduces dimensionality and identifies latent factors
- MANOVA (Multivariate ANOVA) – Compares multiple dependent variables across groups

20. Justify the statement: “Research is concerned with fact-finding, analysis, and evaluation.”

Justification of the Statement

“Research is concerned with fact-finding, analysis, and evaluation.”

Fact-Finding:

- Research begins with collecting accurate and relevant information about a problem or phenomenon.
- This involves gathering primary or secondary data to understand what is already known and what gaps exist.

Analysis:

- After data collection, research involves examining and interpreting the facts.
- Statistical and analytical methods are used to identify patterns, relationships, and trends.

Evaluation:

- Finally, research involves assessing the meaning and implications of the findings.
- Conclusions are drawn, hypotheses are tested, and decisions or recommendations are made based on the evidence.